

ATC quick guide to operating the Transponder Landing System (TLS)

1. The pilot will contact ATC and request TLS approach.
2. ATC must enter the transponder squawk code for the aircraft requesting TLS approach in the ILS box matching the assigned frequency.
 - The squawk code may be entered on the RCU by double clicking on the surveillance track symbol or,
 - The squawk code may be entered by typing on the RCU numeric keypad
3. The code should appear in the "Transponder ID" box for the ILS box matching the assigned frequency shown in Figure 1.
4. As the aircraft is turning inbound, ATC should then press Acquire in the ILS box matching the assigned frequency.
5. The Loc frequency transmitted by TLS is noted within the ILS box e.g. 108.15, glide slope frequency pairing is automatic.
 - Note that if ATC enters a transponder code and presses Acquire in the GCA box (where no loc frequency is indicated), TLS will not transmit ILS guidance and no ILS needle indications will appear in the cockpit.
 - If ATC presses acquire and the aircraft departs the service volume then ATC will need to press acquire again when the aircraft returns to conduct a TLS approach.
 - If the aircraft is outside the service volume when ATC presses acquire, there is a configurable timeout interval (normally 2 minutes) on the acquire cycle.
6. ATC shall authorize one aircraft to use one TLS frequency at a time.
7. ATC may use Acquire in the GCA box to provide PAR guidance to an additional aircraft without using an additional ILS frequency e.g. if equipped with one GTU, then a second aircraft flown by a pilot with PAR currency may land with commands from a PAR authorized approach controller.
8. The GTU must show status color GREEN to control the GTU from this RCU.
9. If the status is the color blue, then this RCU does not have control of the GTU.

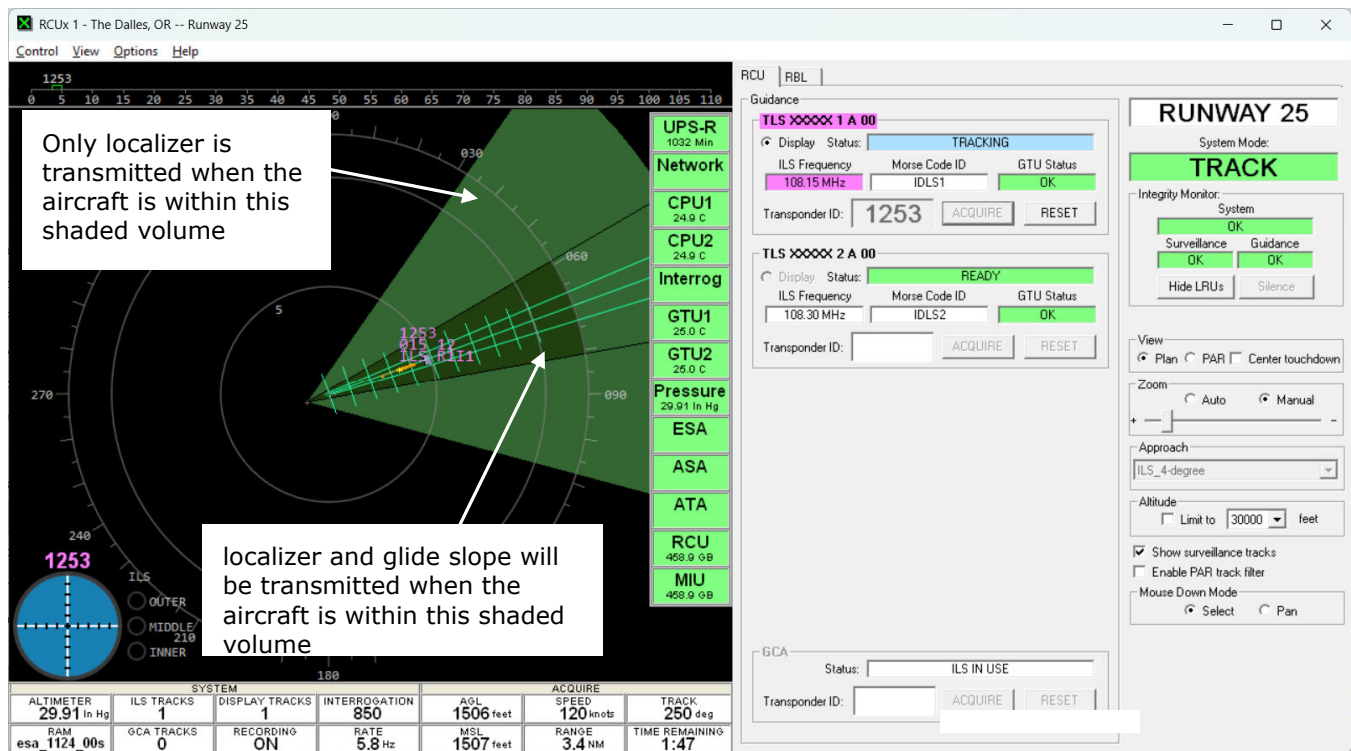


Figure 1 - TLS RCU console

Three Guidance status condition explained:

Ready: The TLS and GTU are operating normally, but not providing guidance to an aircraft

Loc only: The GTU is active and transmitting just the localizer to the aircraft. The service volume range is 25nm and 70 degrees in width (Figure 1).

Tracking: If the aircraft is also within the glide slope volume (Figure 1), and heading inbound then TLS will transmit both the localizer and glide slope. Note that if the aircraft is within the glide slope volume but headed outbound, the glide slope is not transmitted.

Other important information

Frequency: Localizer and glide slope are transmitted on the standard ILS frequency pair that is configured for the TLS.

Timeout: If the aircraft is outside the localizer service volume when "AQUIRE ILS" is selected, then the TLS will remain in search mode for 2 minutes. If the aircraft cannot be detected within the localizer service volume within two minutes, then the TLS will timeout on the search.

Map Rotation: To orient the RCU map to the preferred rotation, press ctrl-alt-left mouse click. Each time this combination is selected, the RCU map will rotate. Press this combination repeatedly until the desired orientation is displayed.

TLS Critical area

When an aircraft is using TLS, ATC must keep aircraft and other ground vehicles clear of the TLS Critical Area and assist aircraft in observing any TLS critical area hold lines associated with taxiways and aprons.

Standard IAP regulations require that the critical area be protected from unrestricted vehicular access regardless of weather conditions whenever an aircraft is conducting a precision approach using the TLS.

If the TLS guidance signal is interrupted:

When the aircraft departs the GS service volume, the GS transmission from TLS will cease. Also, as the aircraft departs the localizer service volume, the localizer transmission will cease, these are both normal interruptions in guidance since in this scenario, the aircraft has departed the service volume. In the case of departing the TLS localizer service volume, the transponder code will clear and the tack state will change to "Ready".

Normal ILS techniques must be used to intercept the glide path from below, advise the pilot of this technique if the guidance was interrupted when intercepting from above.

If there is a red status indication on a TLS component, notify the TLS technician so that the TLS alarm indication can be resolved.

The wingman must set his transponder to standby when flying closer than one nautical mile to the lead aircraft, this is the normal procedure to allow transponder-based systems (TCAS, SSR and TLS) to work effectively.